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B. E. (Fifth Semester) Examination, April-May, 2016

(New Scheme)

(Mech. Engg. Branch)

TURBO MACHINERY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question is compulsory. Solve any two parts from (b), (c) and (d). Use of steam table and Mollier chart is permitted.*

1. (a) Write principle of operation of stem turbine. 2

(b) Find the condition of maximum blade efficiency in single stage impulse turbine. 7

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(c) What is difference between pressure compounding and velocity compounding 7

(d) In a single-stage impulse turbine, the steam velocity at nozzle mouth is 300 m/s. The nozzle angle is 18°s the mean blade velocity is 144 m/s. Draw to a suitable scale the velocity diagram for the steam assuming that the outlet angle of the blade is 3° less than inlet angle and that te relative velocity of steam at outlet from the blade is ·84 of the relative vcelocity at entrance if the power to be developed is 1000 kW. Calculate the mass of steam that must pass through the turbine per sec. 7

2. (a) Define degree of reaction. 2

(b) What is the condition of maximum gross stage efficiency in Parson's reaction turbine? 7

(c) Explain :

(i) Losses due to witness of steam

(ii) Carry over losses 7

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(d) Calculate the maximum theoretical efficiency for a turbine stage having a degree of reaction of ·5 and a velocity coefficient of ·94 when the exit blade angle is 25°. 7

3. (a) What is state point locus? 2

(b) What do you mean by governing of steam turbine? Explain nozzle control governing 7

(c) What is the difference between throttle and by-pass governing? 7

(d) The initial enthalpy at inlet to the nozzles of a turbine stare stare is 2680 kJ/kg and the enthalpy at outlet from the blades is 2590 kJ/kg if the total internal steam flow through the turbine is 10 kg/s and the power absorbed in disc friction is 7·35 kW. What is the workdone by the blading per kg of steam? 7

4. (a) Define gas turbine. 2

(b) State merits and demerits of closed cycle gas turbine over open cycle. 7

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(c) Derive an expression for thermal efficiency in terms of pressure ratio for a simple gas turbine cycle. 7

(d) Air at temperature of 15°C enters a gas turbine plant working at pressure ratio of 15. Turbine inlet temperature is 1250°C polytropic efficiency (small stage efficiency) of compressor and turbine is ·91. Assume $C_p = 1{\cdot}005$ and $1{\cdot}128$ for air and gases respectively and calorific value of fuel used = 42000 kJ/kg of fuel.

Calculate :

(i) overall efficiency

(ii) specific output

(iii) fuel to air ratio

(iv) specific fuel consumption 7

5. (a) Define slip factor for centrifugal compressor. 2

(b) Describe working principle of Axial flow compressor. 7

(c) Give comparison between centrifugal and axial flow compressor. 7

(d) Discuss surging, choking and stalling phenomenon. 7

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